

APPENDIX F: ITD ENVIRONMENT

The Ministry of SaskBuilds and Procurement, Information Technology Division ("ITD"), manages information technology for all of the Government of Saskatchewan ("GoS") ministries.

ITD Mandate

ITD's mandate is to meaningfully lead the development of citizen-centric government services through cross government collaboration and the application of modern technology solutions. The guiding frameworks include:

- A **"Citizen Centric"** focus - Information technology is an investment in developing business and social solutions leveraging technology to improve **service for the citizen**. Investments and solution delivery must keep in mind the eventual impacts and outcomes to the citizen;
- A **"One Government"** approach - Government is a single service, facilitated through cross Ministry collaboration, to support the citizens of Saskatchewan. Information technology investments are viewed as a single orchestrated platform on which to serve the citizen;
- Alignment with the **"Saskatchewan Growth Plan"** - Initiatives are in support of the Growth Plan for the province. The contracts and actions that ITD undertakes, such as IT infrastructure and constructs are the enabling factors for ministry initiatives that fulfill their contribution to the growth plan.

ITD is responsible for architectural and quality standards associated with solution design and implementation. ITD maintains standards to protect the information of our client ministries. Iterative approval processes are in place to review and approve successive levels of detail regarding solution proposals. These include a review of the Proponent's initial high level solution proposal, and subsequent levels of elaboration within the implementation project for conceptual, logical and physical designs.

In support of a One Government, Citizen Centric approach aligned with the Saskatchewan Growth plan, the following Technical Standards and Enterprise Services are required to be considered in any solution proposals. Where applicable, there is a preference towards solutions based on the standards and services described below. Non-standard solutions will be considered, but must be approved through the ITD governance processes and acceptance by the Executive Director of Strategy and Architecture, Information Technology Division, Ministry of SaskBuilds and Procurement.

TECHNICAL STANDARDS

Application Authentication

Government Employee and Partner Authentication (B2B):

- GoS Azure Active Directory

Citizens and Businesses (B2C):

- Saskatchewan Account (Vivvo CitizenOne)

Application Server

Java Workloads

- IBM WebSphere Application Server 9.0
- Oracle WebLogic 12c R2

Microsoft .NET Workloads

- Microsoft IIS 10, 11 (Windows Server 2019)

Audio/Video Meetings - Internal - Point to Point

- Microsoft Teams

Business Productivity Suite

- Microsoft Office 2016
- Microsoft Office 365 Pro Plus

Cloud Services (IaaS, PaaS)

- Microsoft Azure (Canada regions)
- Future: other Canadian region providers

Electronic Payments

- Paysafe (card not present, on-line payments)
- Moneris (point of sale)

Enterprise Collaboration & Document Repository

- Microsoft SharePoint Online
- Microsoft Teams

Enterprise Customer Relationship Management

- Microsoft Dynamics 365 CE Online
- Microsoft Power Apps Portal

Enterprise DBMS

- IBM DB2 v11.1
- Oracle 18c, 19c
- Microsoft SQL Server 2017

Enterprise DevOps Services

- Microsoft Azure DevOps Services
- Microsoft Azure DevOps Boards
- Microsoft Azure Pipelines Managed
- Microsoft Azure Test Plans
- Microsoft Azure Artifacts
- Microsoft Azure Repos Git

Enterprise GIS :

Enterprise GIS – on premises

- Esri ArcGIS Server 10.8.1
- Esri Image Server 10.8.1
- Esri Portal 10.8.1
 - Relational Datastore (Oracle, MS SQL Server)
 - Tile Cache Datastore
- Esri ArcGIS Desktop 10.8.1
- Esri ArcGIS Professional 2.6.2

Enterprise GIS – 3rd Party Solutions

- VertiGIS Geocortex 4.14.x
- VertiGIS Geocortex 5 series installed on-premises
- Safe FME Desktop 2020.2

Enterprise GIS - Esri Solution as a Service

- Esri ArcGIS Online (AGOL)
- Esri ArcGIS Online (AGOL) GeoHUB

Enterprise Reporting

- Cognos Analytics 11.1.x

Enterprise Service Management

- ServiceNow

Enterprise Electronic Signature

- DocuSign

Multifactor Authentication

- Entrust IdentityGuard
- Azure Multi Factor Authentication (Azure MFA)

On-premises Intel Server Virtualization

- VMware ESXi 6.7

Network Protocol - LAN / WAN

- IPv4

Privileged Identity and Access Manager

- Cyberark

Programming Languages / IDE

- Java – Eclipse IDE 2020-19
- Java – IBM RAD
- Java – Oracle JDeveloper 12c
- Java - VS Code
- PHP 7.1, 7.4
- Microsoft Visual Studio 2017, 2019, .NET and Power Platform

Project Management

- Microsoft Project Online

Remote Network Access

- Cisco AnyConnect VPN Client 4.9
- Cisco Clientless VPN
- Microsoft Application Gateway

Server Operating System

- AIX 7.1, 7.2
- Red Hat Enterprise Linux 7.6, 8
- Windows Server 2016, 2019

Server Virtualization

- VMware ESXi 6.7

Mobile OS

- Apple iOS 15, 14
- Android 11.x, 10.x

Tablet Hardware

- Apple iPad

Virtual Desktop Infrastructure

- VMWare Horizon View 7.8
- Windows Virtual Desktop

Web Application Firewall (as a Service)

- Radware WAFaaS

Web Browser

- Microsoft Edge
- Google Chrome
- Mozilla Firefox

Web Server

- Apache HTTP Server 2.4 or Web Server based on Apache HTTP Server 2.4 (IBM HTTP server, for example)
- Microsoft IIS 11, 10
- Microsoft Power Portals

Workstation Anti-Virus

- McAfee EndPoint Security 10.6
- McAfee EndPoint Security 10.5

Workstation Application Delivery

- Citrix XenApp 6.5 – App Streaming
- Microsoft System Center Configuration Manager 1902

Workstation Operating System

- Windows 10

ENTERPRISE APPLICATION STANDARDS

Enterprise Financial Standard

The GoS has adopted Oracle E-Business Suite (Release R12.1.3) as an enterprise solution. The current modules are listed below.

- **Oracle Financials**
 - General Ledger
 - Accounts Payable
 - Accounts Receivable
 - Assets
 - Cash Management
 - CPG Trade Accounting
 - Global Consolidation System
 - iAssets
 - Public Sector Accounts Payable
 - Public Sector Accounts Receivable
 - Public Sector Budgeting
 - Public Sector General Ledger
 - Revenue Accounting
- **Oracle Internet Expense**
- **Oracle Inventory Management**
- **Oracle iProcurement**
- **Oracle iStore**
- **Oracle iRevenue**
- **Oracle Project Billing for Project Costing**
- **Oracle Project Costing**
- **Oracle Purchasing**
- **Oracle Sourcing for Purchasing**

The GoS has current plans for upgrading / migrating its Financial, Procurement and Human Resources solutions to a cloud based solution. As such, any required integrations for current solutions should be through the Enterprise Integration Services described below.

Enterprise Digital Standard

As part of the One Government Strategy, Saskatchewan.ca is the single window for online content and Saskatchewan Account is the enterprise approach for Single Sign On. Saskatchewan.ca and Saskatchewan Account are managed and supported by the Citizen Centric Program Delivery branch within the Ministry of SaskBuilds and Procurement.

See associated Appendix: Developing Public-Facing Online Services for requirements on designing, building and testing public facing online services.

Saskatchewan.ca

The Saskatchewan.ca website must be the single entry point for all content and any digital service offered by the GoS. Ministries will use Saskatchewan.ca as their primary promotional web address and ministries will use Saskatchewan.ca Account for any services where authentication is required for public facing services. Application integration must conform to the Enterprise Integration Standard below.

Saskatchewan Account

Saskatchewan Account (GoS Single Sign On platform) must be used for public (Citizens and Organizations) authentication and profile management for all public facing services.

- Saskatchewan Account platform is provided by Vivvo Application Studios CitizenOne Product
- Leverages SAML2.0 for Authentication

Application integration must conform to the Enterprise Integration Standard below.

Enterprise Collaboration Standard

The Government of Saskatchewan has adopted Microsoft SharePoint Online as the standard for team collaboration. This service is used to manage documents as well as team communications. At present OneDrive for Business and Microsoft Teams are added to supplement team collaboration.

Enterprise Customer Relationship Management Standard (eCRM)

The Government of Saskatchewan (GoS) has adopted the Microsoft Dynamics 365 Online and the PowerApps platform as the standard in setting up GoS Ministries for managing Government's interactions with citizens and businesses of the province and beyond. Staying connected to the customers and streamlining of processes are achieved through the design and implementation of enterprise standards for customer centric business model, customer relationship management, customer engagement and intake, automation of business processes, contact management, case management, grants management, contract management, data analysis and compilation of customer data, integration with other GoS enterprise systems and real time dashboards.

Enterprise CRM includes built in integration with SharePoint Online, Exchange Online, Azure Active Directory and other Office 365 business application stack that are provisioned within the GoS Office 365 and Azure tenancy. This standard includes the Microsoft provisioned Common Data Model, Common Data Service, PowerBI, PowerApps and PowerPortals. This standard is provisioned from Microsoft's Canada Central data center and this standard is certified to host Class A solutions. The content portal is served through GoS Enterprise Integration Standard.

Enterprise Geographical Information Standard (eGIS)

The Government of Saskatchewan utilizes the Esri Enterprise GIS suite of products as the standard for mapping and geospatial information. The Enterprise eGIS deployment enables the government to manage and maintain information in either standard spatial data formats – vector or raster content. Current solutions can also enable temporal (time) variables to allow modeling and predictive analytics.

The Esri eGIS Enterprise Solution provides GoS with the ability to access, update and contribute content from desktop, mobile devices (Android/iOS/Windows) or IOT solutions. Current versions of products are outlined in the "Technical Standards – Enterprise GIS" section above.

Current provincial standards follow ISO and industry standards (**ISO 19139 / ISO 19115**) for Federal Geographic Data Committee (FGDC) metadata and XML encoding. Spatial data standards for Canada are currently transitioning to **North American Profile**.

Current Projection Standard for Saskatchewan is **NAD 1983 CSRS UTM Zone 13N**.

Interoperability solutions in the enterprise enable the government to accept spatial data in multiple formats. Location formats include: Civic Address; Geographic Coordinates; Universal Transverse Mercator (UTM) Coordinates; Land Description; Survey Plan; ISC Parcel Number; and River Lot. Preferred file formats include Esri standards (shapefile SHP, geodatabase GDB). Additional published ArcGIS data interoperability supported formats can also be considered - <https://www.esri.com/en-us/arcgis/products/arcgis-data-interoperability/supported-formats>.

The eGIS environment follows the GoS standard to support data exchange through **REST** Services. The preferred API authorization is through OAuth 2.0 tokens. Secure file transfers are also supported. Other standards information relevant to the enterprise implementations can be found on the Esri's site - <https://www.esri.com/en-us/arcgis/open-vision/initiatives/standards-specifications>.

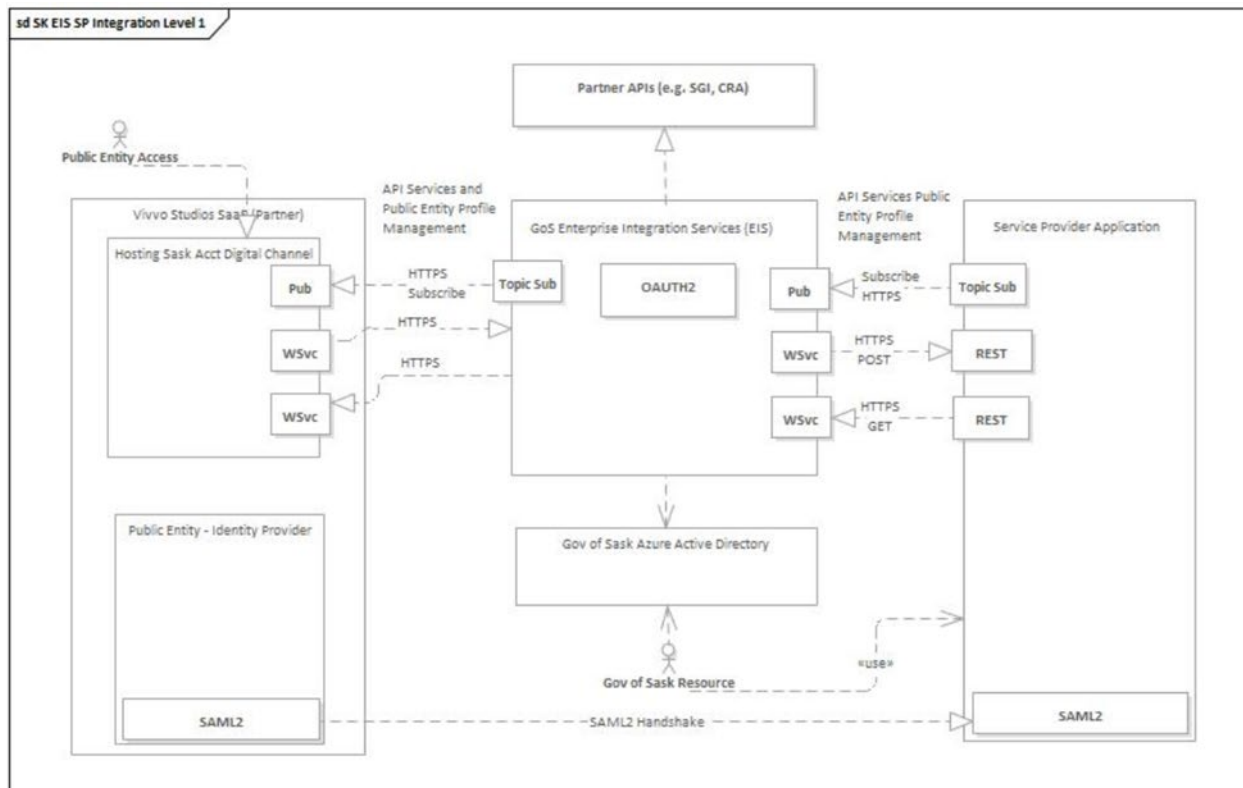
Esri and the Government of Saskatchewan are following and using the Open GIS Standards from the Open Geospatial Consortium - <https://www.ogc.org/docs/is>.

Enterprise Integration Services

Integration between Government of Saskatchewan applications or services that exchange data (regardless of where they are hosted) are facilitated through the GoS Enterprise Integration Services. These services are delivered using the Microsoft Azure Integration Service (AIS) platform. The Enterprise Integration Standards are further explained below.

Enterprise Integration Standard

The Government of Saskatchewan integration standard is to support data exchange through REST web services. This standard is a GoS ITD developed Integration platform that provides centralized service enabling technologies to securely deliver Information Assets (Data) among collaborating systems to address the business needs of the Government of Saskatchewan. Examples of collaborating systems include Public facing web browsers, partner systems (SGI, CRA), SaaS as in Enterprise Customer Relationship Management, PaaS/SaaS – Saskatchewan.ca Account, on-premises systems including legacy (CRM2016, MIDAS), etc. The high level integration framework is depicted in the diagram that follows:



Enterprise Reporting Standard

The Government of Saskatchewan has adopted Cognos Analytics 11.1.x as the standard for reporting and business intelligence. Enterprise Reporting is preparing GoS Ministries in moving up the analytics maturity curve to predictive capabilities through the design and implementation of enterprise standards for data warehousing, data analysis, real time dashboards, content packs, embedded models and presentation technologies as it relates to Data Analytics. Enterprise Reporting can take many different approaches. Power BI Online is minimally used for GoS Microsoft Power Platform based solutions.

Enterprise Search Standard (Web Content)

The Government of Saskatchewan has adopted Coveo Enterprise Search 7.0 as the standard for searching enterprise web content.

If the solution has built-in search functionality it may leverage that search tool. If the solution does not have search the project will first evaluate the enterprise online search tool, Coveo as a way to meet the requirements prior to using a new search tool. A business case to deviate from the enterprise tool, will need to be proven prior to a different search technology being implemented.

Enterprise Operational Reporting Service (eORS)

The Government of Saskatchewan has adopted Enterprise Operational Reporting Service (“EORS”) as a platform to host managed reports, Ministry ad-hoc reports and data for the GoS. It consists of a central data warehouse, extract/transform/load (“ETL”) services, business intelligence and analytics report engine and associated processes to ensure scalability, security and availability of the platform service.

IT Service Management

The Government of Saskatchewan has adopted the ServiceNow platform for IT Service delivery (incident, problem, change, release, and knowledge management). In support of a “one number to call” approach for providing IT support to Government employees, integration between GoS service management processes and systems with associated vendor services should be considered. This integration should be through the Enterprise Integration Services described above.

Electronic Payments

The Government of Saskatchewan currently utilizes Paysafe for card not present (on-line) payments, and Moneris for point of sale transactions. Note – all applications requiring electronic payments will be through direct integration with Paysafe to maintain PCI compliance.

Application Lifecycle Management

Applications and solutions requiring custom development and deployment of code for the Government of Saskatchewan are managed utilizing the GoS Microsoft Azure DevOps platform. This includes, but is not limited to product backlog, user stories, source, configuration, release, testing and deployment management.

Technology Lifecycle Policy

ITD mandates an N-1 policy for solution acquisition and support. Specifically, current product (N) and the previous product version (N-1) are to be considered for implementation. Plans for keeping solutions up to date must be included in any proposals. The N-1 support policy enables only solutions based on the newer products which are supported under the product vendor’s Mainstream or Standard support. Older (N-2) products and any product where Mainstream support has ended, are not entitled for consideration.